AI-POWERED CPS FOR SKIN DISEASE DETECTION

TEAM NAME: THE_CRACKED_CODERS

THEME: Health: AI-Powered CPS for Skin Disease Detection

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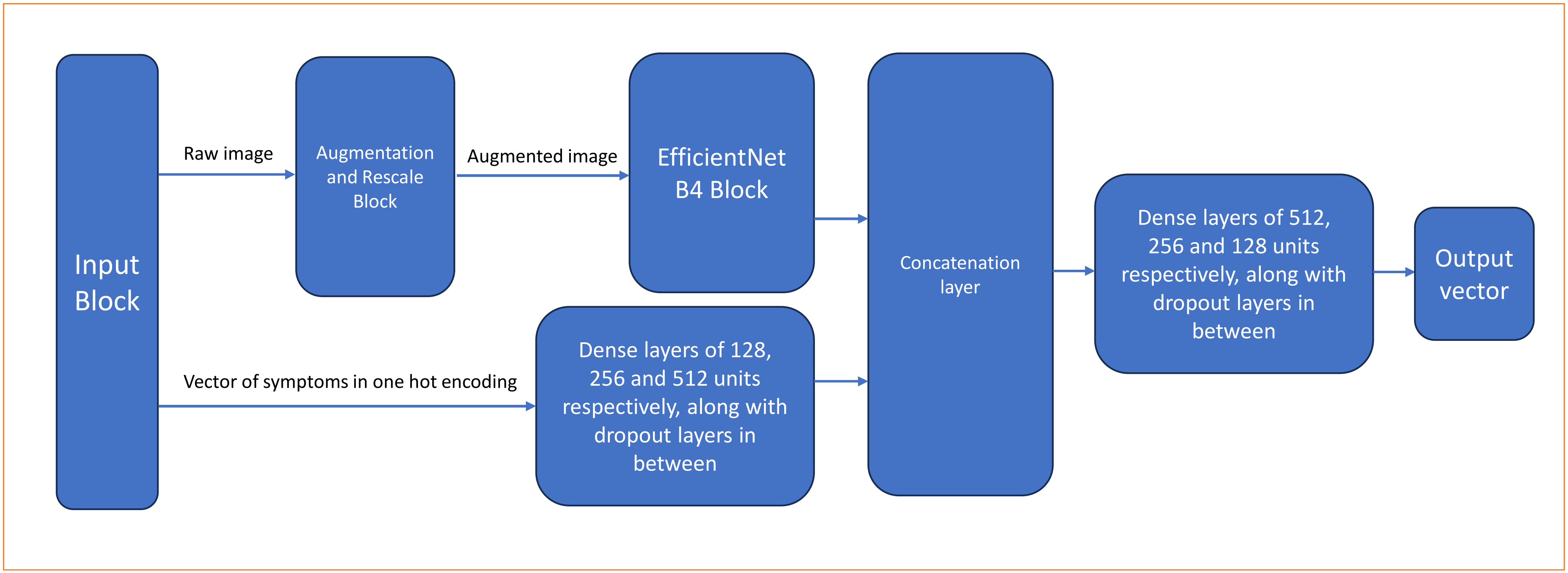
HEALTH: AI-POWERED CPS FOR SKIN DISEASE DETECTION

- Skin diseases are among the most common health problems globally.
- Early diagnosis for serious conditions is of utmost importance.
- Access to dermatologists is quite limited in remote areas for early diagnosis.
- To tackle this, we have created an AI powered CPS for skin disease detection.
- This helps to eliminate the problems of early diagnosis and risk categorization.

OVERVIEW OF AI-POWERED CPS FOR SKIN DISEASE DETECTION

- This is an image classification system for skin disease detection.
- It was trained on the image dataset for 27 skin diseases, e.g.- Melanoma, Hives.
- It requires an image of the skin and the symptoms(prompted by a predetermined set of questions which needs to be answered) as input.
- It will generate the name of the skin disease among those 27 diseases as output.

SYSTEM ARCHITECTURE



Architecture of the image classification model

Workflow:

- **Input Data:** The process begins with two types of input data: a "Raw image" and a "Vector of symptoms in one hot encoding".
- **Image Preprocessing:** The "Raw image" is first sent to the "Augmentation and Rescale Block".
- **Image Processing:** The resulting "Augmented image" is then processed by the "EfficientNet B4 Block".
- **Symptom Processing:** Simultaneously, the "Vector of symptoms" is processed through a series of "Dense layers of 128, 256 and 512 units respectively, along with dropout layers in between".
- **Feature Concatenation:** The outputs from both the EfficientNet B4 Block and the symptom dense layers are combined in the "Concatenation layer".
- **Combined Feature Processing:** The concatenated data is then further processed by another set of "Dense layers of 512, 256 and 128 units respectively, along with dropout layers in between".
- **Final Output:** The final step is the generation of the "Output vector".

DATASET USED

- This was a custom made dataset made by combining multiple datasets from many different sources like Kaggle, HAM10000 and DermNet, Roboflow, etc.
- Size of the dataset: 14.67 GB.
- Source of the dataset: https://kaggle.com/datasets/bc094de636703778a683389d5a2d60ca848254baed5ad2c7f93700a5cd0840dd
- Preprocessing steps include Flipping, Zooming, Changing Brightness, Translation, Changing contrast, Changing Saturation, Changing Hue, Rotating and Rescaling.

Melanoma

Chicken Pox

Eczema

Monkeypox

MODELS USED

AI/ML/DL Techniques Applied:

- Architecture: Multi-Modal Hybrid Network (CNN + ANN).
- Visual Stream: Uses EfficientNetB4 which utilizes Convolutional Neural Network (CNN) for feature extraction from dermoscopic images.
- Clinical Stream: Dense Neural Network for processing the symptom matrix.
- Regularization: Dropout layers and Data Augmentation to prevent overfitting.

Justification of Model Choice:

- Usage of EfficientNetB4: Outperforms ResNet50 with higher accuracy and fewer parameters, optimizing processing for high-resolution images. It simultaneously scales resolution (380x380), depth, and width to capture fine-grained lesion textures missed by traditional CNNs.
- Holistic Diagnosis: Integrating visual data with clinical symptoms improves sensitivity and specificity compared to image-only models.
- Robustness: The architecture is designed to handle the high variability in skin tone and lighting conditions present in the dataset (HAM10000/DermNet).

IMPLEMENTATION DETAILS

Hardware/software setup:

- Used the Kaggle editor for designing the model.
- Used the NVIDIA Tesla P100 GPU for training and testing the model, present in the Kaggle editor.
- Used our local machine for testing the integration of web API with our model by hosting it locally.

Programming Tools & Frameworks:

- TensorFlow: Backend for building and training the EfficientNetB4 and ANN models.
- Flask is used to create the web API that serves the model predictions to the user interface.
- Pandas: For handling the structured symptom matrix and CSV datasets.
- NumPy: For efficient matrix operations and numerical computations.
- PIL (Python Imaging Library): For loading and resizing images before prediction.
- Scikit-learn: Used for evaluating model metrics like Accuracy, Precision, Recall and F1 score.

RESULTS-QUANTITATIVE

Evaluation Metrics:

- Overall Accuracy: 96.08%, indicating highly reliable classification across 27 distinct disease classes.
- Weighted F1-Score: 94.88%, demonstrating a robust balance between precision and recall despite significant class imbalance.
- Key Successes: Achieved 100% Precision and Recall for critical conditions like Melanoma (most dangerous skin cancer) and viral infections (Chickenpox, Monkeypox).
- Areas for Improvement: Identified performance drops in "tail" classes like Atopic Dermatitis (Precision: 0.00) due to visual similarity with Eczema and low data support.

Baselines and Comparison:

- Architecture Efficiency: Selected EfficientNetB4 over traditional models like for superior parameter efficiency and accuracy.(Approximately 7% in terms of Top-1 accuracy)
- Multi-Modal Advantage: Integrating clinical symptoms overcomes "visual ambiguity" (e.g., distinguishing Eczema from Psoriasis), significantly reducing misdiagnosis compared to image-only models.

CLASSIFICATION REPORT

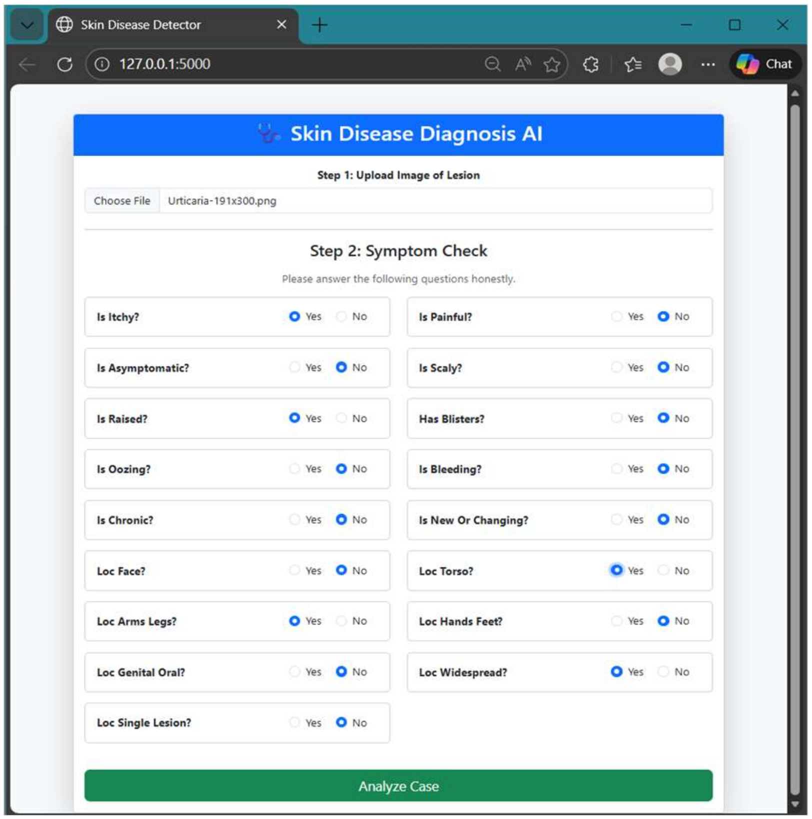
Class	Precision	Recall	F1-score	Support
Acne or Rosacea	1.00	1.00	1.00	312
Actinic Keratoses	1.00	1.00	1.00	88
Atopic Dermatitis	0.00	0.00	0.00	65
Basal cell carcinoma	1.00	1.00	1.00	2658
Benign keratosis-like lesions	0.39	1.00	0.56	263
Chickenpox	1.00	1.00	1.00	113
Cowpox	1.00	1.00	1.00	99
Dermatofibroma	1.00	1.00	1.00	25
Eczema	0.63	1.00	0.78	112
HFMD	1.00	1.00	1.00	242
Healthy	1.00	1.00	1.00	171
Herpes HPV and other STDs	1.00	1.00	1.00	102
Lupus	1.00	1.00	1.00	34
Lyme Disease	1.00	1.00	1.00	95
Measles	1.00	1.00	1.00	83
Melanocytic nevi	1.00	1.00	1.00	6383
Melanoma	1.00	1.00	1.00	335

Class	Precision	Recall	F1-score	Support
Monkeypox	1.00	1.00	1.00	426
Psoriasis Lichen Planus and related diseases	1.00	1.00	1.00	88
Scabies	1.00	1.00	1.00	70
Seborrheic Keratoses and other Benign Tumors	0.00	0.00	0.00	411
Squamous cell carcinoma	1.00	1.00	1.00	64
Tinea Ringworm Candidiasis and other Fungal Infections	1.00	1.00	1.00	102
Urticaria Hives	1.00	1.00	1.00	53
Vascular Lesions or Vascular Tumors	0.00	0.00	0.00	26
Vitiligo	1.00	1.00	1.00	82
Warts Molluscum and other Viral Infections	1.00	1.00	1.00	289

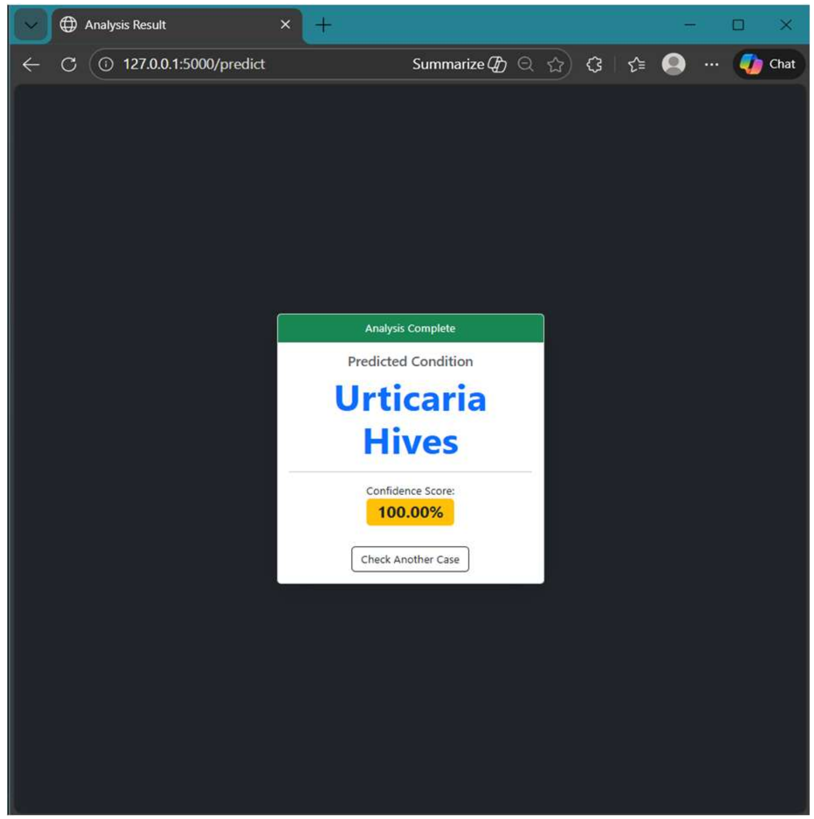
Overall Metrics:

Accuracy: 0.9608, Precision: 0.9430, Recall: 0.9608, F1 Score: 0.9488

RESULTS-QUALITATIVE



UI for uploading image and answering the symptoms



Displaying the result after clicking “Analyze Case”

Image used for the classification (Urticaria Hives)

DEPLOYMENT PLAN

Cloud & Web-Based CPS Integration:

- Centralized Cloud Access: Deployed as a modular Flask web app, ensuring the diagnostic tool is accessible on any device via browser without installation.
- Universal Interface: Simple HTML/Bootstrap frontend allows users to easily upload images and input symptoms via a responsive questionnaire.

Real-World Feasibility:

- Hardware Agnostic: Cloud-side processing offloads heavy EfficientNetB4 inference, making the system functional even on low-end smartphones.
- Scalable Architecture: Decoupled backend design allows independent scaling to handle high traffic volumes.
- Future-Ready: "Client-neutral" API design enables seamless integration with future mobile apps (React Native/Flutter) for broader reach.

CHALLENGES FACED

Challenges faced:

- Finding the right platform to train the model.
- Gathering images for certain disease classes like Dermatofibroma, Lupus, Scabies, etc.

Overcoming the challenges:

- Used Kaggle editor for training the model using powerful GPUs.
- We used image augmentation techniques like Flipping, Zooming, Changing Brightness, Translation, Changing contrast, Changing Saturation, Changing Hue, Rotating and Rescaling to compensate for the lack of images for these classes.

FUTURE WORK & SCALABILITY

- Increase the number of supported skin diseases by adding additional datasets, allowing the system to cover a broader scope.
- Division of the Search Engine into Two Intelligent Modes:

  A. **Quick Diagnosis Mode** : Uses EfficientNetB0 and ANN(for accepting symptoms) for faster classification by combining image inputs with a predefined set of symptom-based questions.

  B. **Deep Analysis Mode** : Uses EfficientNetB4 for image recognition along with predefined symptom inputs. Additionally, NLP techniques can be applied to extract extra disease symptoms from free-text user inputs.
- Implement secure encryption standards such as **JSON Web Encryption (JWE)** to ensure privacy and confidentiality of sensitive medical images along with user data.
- Improved UI/UX and Performance Optimization like lowering the **LCP (Largest Contentful Paint)**.
- Implement Database and Backend servers and cloud deployment for high accessibility.